# Defending Earth: Asteroid Reconnaissance

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**Abstract**

Using the 0.5-meter optical telescope at Anderson Observatory[1], we captured images and logged the positions of three main belt asteroids; asteroids 75 Eurydike, 366 Vincentina, and 33 Polyhymnia. 1943 Anteros, a near earth object, was also imaged and logged. Three 60-second exposure images were taken of each asteroid, with 30 minutes between each image to determine movement by finding the changing position of each object. Image calibration was completed using nine dark, flat, and bias frames to remove most pixel interference using the Siril software.

## 1 Introduction

Asteroids contain debris left over from the formation of the solar system [1]. Main belt asteroids exist between Mars and Jupiter. When objects are brought closer to Earth due to planetary gravity, they become Near Earth Objects (NEO). t [2]. Asteroid and Near Earth Object (NEO) data is primarily stored and maintained in the Minor Planet Center and NASA's JPL Small-Body DataBase.

## 2 Observations

The 0.5-meter optical telescope at the Anderson Observatory was used to observe the asteroids. This telescope is equipped with an STT-8300M CCD Camera, which has a frame of $3326 \times 2504$ pixels and is controlled by the MaxIm DL software. The pixel size of the CCD is 5.4 $\mu$m$^2$. The focal length of the telescope is 3.543 m, the field of view of the telescope is $\approx 17.9' \times 13.1'$ and the pixel area is $\approx 0.33''$.

Three 60-second exposure light frames were taken of each asteroid, with roughly 30 minutes between each frame to allow for the objects to move in relation to the background stars. For image calibration, Nine 60-second dark frames were taken, in addition to nine biases and nine flats.

## 3 Data Reduction

Our images were processed using the Siril Program. Siril is an astronomical image processing tool that is capable of calibrating images and taking the median of a set of darks, flats, and biases to produce master frames. First, the Mono Prepossessing script was used to calibrate the light frames of each of our asteroids. This was done by first median stacking the dark, bias, and flat frames to produce the master frames, subtracting the master bias and dark from every light frame, and dividing by the master flat to produce calibrated luminance images to use in our analysis of the asteroids.

Once the images were calibrated, we did astrometric solving on each of the images. That is, each image was rotated to align the stars in the other images and reflected across a vertical axis to match with the GAIA DR3 catalog. After which, they were converted to 16 Bit images and saved for further analysis in the Astrometrica software.

Astrometrica is an interactive software tool used for astrometric data reduction of CCD images containing minor solar system bodies such as asteroids, comets, and dwarf planets. It is capable of reading in FITS files, automatically referencing stars from the image to a star catalogue, and identifying objects within the image.

[1]Nicholas R. Anderson Observatory, is located on Old Mill Road in Blacksburg, VA and is owned by Virginia Tech

Finally, it is also capable of compiling reports of objects tracked across multiple images, which is what we used to complete our analysis.

We identified the asteroid in each of the four sequences of images by looking for objects that moved with respect to the stars in the background. Once they were identified, the object was selected in each of the images in the sequence in Astrometrica and its object designation, magnitude, location, and other characteristics were saved. This was done for each of the four sequences of three images to end up with 3 data entries for each of the four objects, with each data entry being separated by an observation time of roughly 30 minutes.

# 4 Results

The 4 asteroids observed included 3 main belt asteroids 75 Eurydike (Fig. 2), 366 Vincentina (Fig. 3), and 33 Polyhymnia (Fig. 4) along with 1 near earth object (NEO) 1943 Anteros (Fig. 1). Each asteroid was imaged 3 times over time to track their orbits. The uncertainties in asteroid position determinations come from instrumental precession and determinations of position for the asteroids at the time they were observed, as well as sky clarity. The NEO contains more uncertainty in its measurement due to its proximity to Earth and faster movement across the sky.

In future experiments it would be beneficial to observe the asteroids for a longer period of time to help decrease uncertainty and maybe observe them over a period of days.

## References

[1] Donald K. Yeomans. "Why study asteroids?" In: *Solar System Dynamics* (1998).

[2] Lyle Tavernier. *How NASA Studies and Tracks Asteroids Near and Far*. 2024.

| permID | obsTime | rmsRA | rmsDec | Mag | logSNR |
|---|---|---|---|---|---|
| 33 | 2026-04-03T01:15:45Z | 0.135 | 0.096 | 14.41 | 2.57 |
| | 2026-04-03T01:42:09Z | 0.126 | 0.072 | 14.42 | 2.53 |
| | 2026-04-03T02:11:59Z | 0.072 | 0.073 | 14.33 | 2.56 |
| 75 | 2026-04-03T00:55:28Z | 0.126 | 0.113 | 14.51 | 2.50 |
| | 2026-04-03T01:31:01Z | 0.182 | 0.093 | 14.50 | 2.53 |
| | 2026-04-03T02:04:05Z | 0.046 | 0.047 | 14.58 | 2.55 |
| 366 | 2026-04-03T01:10:25Z | 0.152 | 0.109 | 14.14 | 2.63 |
| | 2026-04-03T01:36:39Z | 0.103 | 0.096 | 14.14 | 2.63 |
| | 2026-04-03T02:07:12Z | 0.044 | 0.056 | 14.12 | 2.64 |
| 1943 | 2026-04-03T01:20:51Z | 0.177 | 0.179 | 14.20 | 2.44 |
| | 2026-04-03T01:45:24Z | 0.135 | 0.117 | 14.31 | 2.44 |
| | 2026-04-03T02:15:32Z | 0.114 | 0.121 | 14.23 | 2.43 |

Table 1: Snap Shot of ADES Report

Figure 1: 1943 Antares Before(left) & After(right). The FOV of the image is 2.6’×2.5’

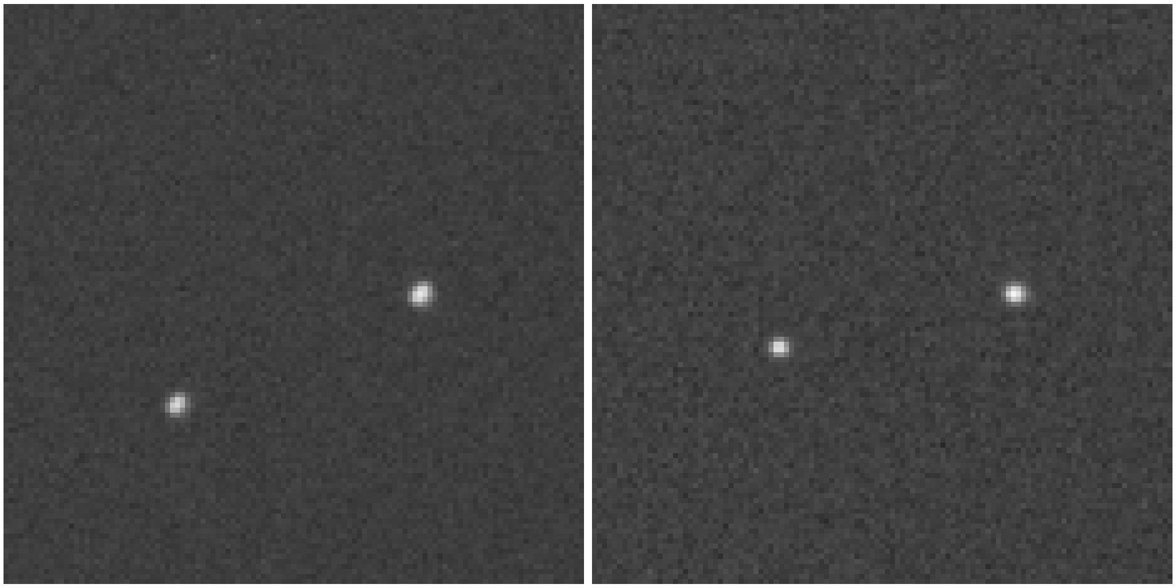

Figure 2: 75 Eurydike Before(left) & After(right). The FOV of the image is 2.3'×2.2'

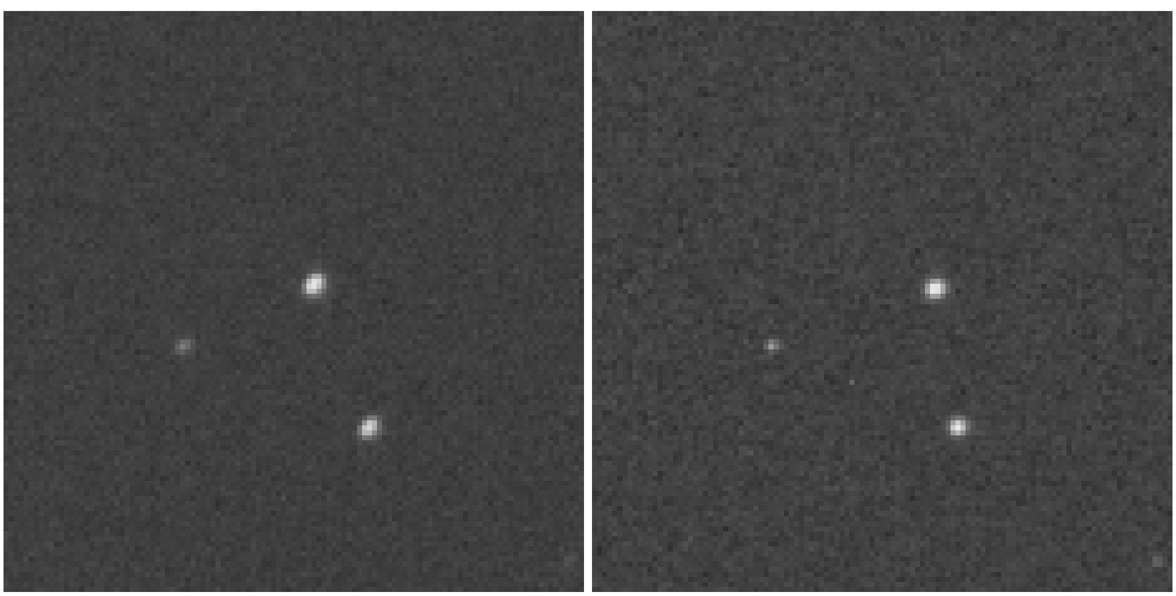

Figure 3: 366 Vincentina Before(left) & After(right). The FOV of the image is 2.3'×2.2'

Figure 4: 33 Polyhymnia Before(left) & After(right). The FOV of the image is 5.9'×5.75'